

Indian Institute of Technology Patna
Mid Semester Examination
Course Name: Deep Learning
Course Code: MA4114/MA472
Date: 26-9-2024

Maximum Marks: 40

Time: 2 Hours

1. Tick whether the following statement is True or False -
"One epoch of Stochastic gradient descent is similar to one iteration of vanilla gradient descent".

[Marks = 1]

- (a) True
(b) False

2. Which of the following statements is true for 1×1 convolution and max pooling in convolutional neural networks (CNNs)?

[Marks = 1]

- (a) 1×1 convolution averages the spatial feature values, while max pooling selects the depth-wise maximum feature values.
(b) 1×1 convolution enhances translation invariance of the network, while Max pooling reduces the impact of affine transformation on the input.
(c) 1×1 convolutions reduce the depth of feature maps, while max pooling reduces spatial dimensions by selecting maximum values.
(d) 1×1 convolution uses less number of parameters than maxpool operation.

3. Compute the cross-entropy loss for a probability prediction vector $[0.7, 0.2, 0.1]$ on a target vector of $[1, 0, 0]$. [Marks = 1]

4. Which one is the derivative of the ReLU (Rectified Linear Unit) activation function?

[Marks = 1]

- (a) ReLU is a non-differentiable function.
(b) 0 for all values of x .
(c) 1 for $x > 0$, 0 for $x < 0$.
(d) 1 for all values of x .

5. Suppose a certain fully connected neural network (FCNN) has a fully connected (FC) layer followed by a dropout layer. The output dimension of the FC layer is 100 and the dropout layer has the probability set to 0.3. What is the number of active neurons of the FC layer during training?

[Marks = 1]

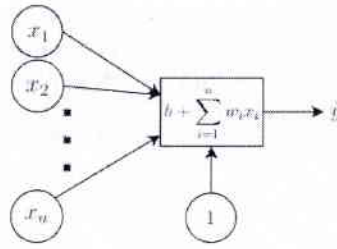
6. Suppose you are convolving an image using a square filter W . The convolution operation on each pixel computes the feature value ($\hat{I}(x, y)$) of pixel $I(x, y)$ using the following formula. What is the height/ width of the filter W ? Need NOT to show computation.

[Marks = 1]

$$\hat{I}(x, y) = \sum_{i=-3}^3 \sum_{j=-3}^3 I(x-i, y-j) * W(i, j)$$

7. Following is a tiny neural network. What kind of decision boundary can this model generate?

[Marks = 1]



- (a) The decision boundary is linear as there is only a single neuron in a single layer in this network.
- (b) The decision boundary is linear as the output neuron does not apply a non-linear function on the linear transformation of the input.
- (c) Irrespective of the size a neural network always generates a non-linear decision boundary.
- (d) There is too little information to tell anything about the nature of the decision boundary of this network.

8. Consider a neural network layer with Batch Normalization (BN) applied after it. If the layer has N input features and M output features, how many additional parameters are introduced by Batch Normalization in this layer? Assume that the Batch Normalization layer includes learnable scale (γ) and shift (β) parameters, and that there are no additional learnable parameters introduced by other aspects of the network. [Marks = 1]

- (a) N
- (b) $2M$
- (c) NM
- (d) $2N$

9. Arrange the following steps in the correct sequence for using the gradient descent algorithm: [Marks = 2]

- (a) Calculate the error between the actual value and the predicted value.
- (b) Reiterate until you find the best weights of the network.
- (c) Pass an input through the network and get values from the output layer.
- (d) Initialize the weights and biases of the model (network).
- (e) Go to each neuron that contributes to the error and change its respective value to reduce the error.

10. In a Convolutional Neural Network (CNN), which of the following statements is true regarding the role of pooling layers? [Marks = 1]

- (a) Pooling layers are responsible for increasing the spatial dimensions of the input image.
- (b) Pooling layers perform convolution operations to detect edges and textures.
- (c) Pooling layers reduce the spatial dimensions of the input while retaining important features.
- (d) Pooling layers are used to introduce non-linearity into the network.

11. (a) Derive the gradient of the Sigmoid and ReLU activation functions. [Marks = 3]

(b) Discuss the impact of these gradients on the training of deep networks, particularly focusing on the vanishing gradient problem for Sigmoid and the exploding gradient problem for ReLU.

[Marks = 2]

12. (a) Derive the update rules for Stochastic Gradient Descent (SGD) with momentum and the Adam optimizer. Show all mathematical steps. [Marks = 4]

(b) Given a cost function $J(\theta)$ with gradient $\nabla J(\theta)$, compare the convergence speed and stability of these optimizers in terms of their parameter updates. Discuss how each optimizer adjusts its learning rates. [Marks = 2]

13. Consider a simple fully connected neural network with one hidden layer. The input is a 3-dimensional vector $\mathbf{x} = [x_1, x_2, x_3]$, and the network has 2 hidden units with ReLU activations. The output is a single scalar y . The network parameters include weight matrices $W_1 \in \mathbb{R}^{2 \times 3}$ and $W_2 \in \mathbb{R}^{1 \times 2}$, and bias vectors $\mathbf{b}_1 \in \mathbb{R}^2$ and $b_2 \in \mathbb{R}^1$.

The output of the network is given by:

$$z_1 = W_1 \mathbf{x} + \mathbf{b}_1$$

$$a_1 = \text{ReLU}(z_1)$$

$$y = W_2 a_1 + b_2$$

Given the loss function $L = \frac{1}{2}(y - \hat{y})^2$, where $\hat{y}$ is the true label, derive the gradient of the loss function with respect to the parameters W_1 , W_2 , $\mathbf{b}_1$, and b_2 . [Marks = 6]

14. Consider a convolutional neural network (CNN) with the following layers:

- Input layer: $32 \times 32 \times 3$ (32x32 RGB image)
- Convolution layer 1: 16 filters of size 3×3 , stride 1, padding 1
- Max-pooling layer: 2×2 filter, stride 2
- Convolution layer 2: 32 filters of size 3×3 , stride 1, padding 1
- Fully connected layer: 128 output neurons
- Output layer: 10 classes

Calculate the total number of parameters (including both weights and biases) and the total number of FLOPs (floating-point operations, considering only multiplication operations) required by the network. [Marks = 6]

15. What are the key differences in the architectural designs of AlexNet, VGG, ResNet, and GoogleNet, and how do these differences impact their performance and efficiency in image classification tasks? [Marks = 3]

16. How does weight initialization affect the overall training of a neural network, and can you explain its impact with an example? [Marks = 3]

